

Response to Reviewers

1. General Response

We thank the Associate Editor and all reviewers for the careful second-round assessment. Three concerns recurred across the reports. These were the theoretical basis of the functional decoupling, the absence of statistical validation, and evaluation on a single benchmark. Each is answered with new analysis or new experiments in the manuscript rather than with rewriting. The table below maps every concern to the change and its location.

Concern	What we changed	Where
Theory of the decoupling (R2, R4)	Exact entropy identity replacing the previous approximate risk decomposition, one derivation per module, and a 2×2 interaction experiment testing their prediction	Sec. 3.5, and Sec. 4.3.2 with Table 6
Statistical validation (R2, R6)	Five seeds, 95% t -intervals, paired t -tests against every compared method with Holm correction	Sec. 4.2.1 and Table 1, with full statistics in Table R1 of this letter
Single benchmark (R2, R4)	Cross-dataset study on Flickr-LDL and Emotion6, placed in the manuscript itself	Sec. 4.5, Table 14
Recent baselines (R4, EiC)	Two 2026 intent methods added and compared, one under a matched backbone, and four recent MLLMs evaluated zero-shot	Sec. 2, Tables 1, 2, 13
Abstract comparability (AE)	Absolute values replaced with relative gains over the strongest prior result	Abstract
Proofreading, bibliography (R4, EiC)	Venue fields corrected, new works discussed individually, 51 items	Bibliography, Sec. 2

Two points of context apply throughout, so we state them once here rather than repeating them under each comment.

The cross-dataset study. We first explored a transfer evaluation on EMOTIC. Its annotation protocol does not naturally match our soft multi-label setting, so we replaced it with Flickr-LDL and Emotion6, which carry native subjective label distributions and are evaluated under standard published protocols. This is why EMOTIC no longer appears in the manuscript.

What this letter adds. The manuscript is at the 35-page limit, so three items live here instead. These are the complete per-method significance statistics (Table R1), the seed dispersion behind the cross-dataset table (Table R2), and the per-class F1 attribution behind the manuscript’s one-sentence interpretability summary (Table R3). Everything else we describe below is in the revised manuscript, marked in red.

2. Response to Reviewer 2

We thank the reviewer for acknowledging the improvements in methodological detail, evaluation, and presentation, and for stating the remaining concerns so precisely. All four are addressed with changes in the manuscript, summarized below.

Comment 1

Nevertheless, there are still some serious scientific concerns. There is still a lack of theoretical justification of functionally separate two types of ambiguity.

Response: *Sec. 3.5 has been rebuilt around an exact identity plus one derivation per module, and Sec. 4.3.2 adds an experiment that tests what they predict.*

We agree, and have rebuilt the analysis on an exact footing. Sec. 3.5 now derives the exact identity $H(Y_c | Z) = H(Y_c | X) + I(Y_c; X | Z)$ (Eq. (15)), which separates irreducible annotator disagreement from label-relevant information the representation discards. Each module is then characterized on its own terms. SLR-C is stated through the conditional mutual information it supplies (Eq. (16)) and the margin it shifts inside the screened candidate set (Eq. (17)). UTD is stated as learning using privileged information, whose mean-squared-error-optimal weight decreases in annotator agreement and so motivates the gate $1 - \omega$ (Eqs. (18)–(19)).

It establishes that the two ambiguities are functionals of different objects, the annotation conditional and the representation map, and therefore admit corrective mechanisms of different type. Whether that pays off is empirical, so we added an experiment to test it. The new 2×2 stratification below confirms the prediction on the semantic axis, where SLR-C gains only in the high-semantic-ambiguity cells, and shows UTD inert wherever annotator agreement is high, with its only significant gain where both ambiguities coincide. The full model does not dominate the doubly-ambiguous cell, so we claim the predicted localization of the two effects rather than a gain from composing them.

Where and what we changed. the derivations (15)–(19) and the module mapping in Sec. 3.5, the 2×2 interaction experiment in Sec. 4.3.2 and Table 6, and the unified-versus-decoupled control in Table 5.

Comment 2

In spite of the inclusion of new experiments, statistical validation is still missing. Confidence intervals, hypothesis testing, and robustness to multiple training runs are necessary to prove that reported improvements are statistically significant.

Response: All three elements are now in place, with the complete per-method statistics in Table R1 of this letter.

We extend the main experiments from three to **five** independent seeds, re-training FDIL and every matched-protocol method on each seed. Table 1 now reports 95% t -intervals instead of bare standard deviations, and Sec. 4.2.1 reports paired two-sided t -tests against each compared method, paired on the training run and Holm-corrected within the family of methods compared on that metric. This is complemented by the sample-level paired bootstrap behind Table 10, so run-to-run and sample-to-sample variance are both quantified.

Outcome. Avg. F1, mAP, and hard-subset F1 are significant against every compared method after correction. Exactly one contrast is not, and we report it as well. That contrast is Samples F1 against our own UTD-only ablation ($p_H = 0.0755$). Because the manuscript is at the page limit, it summarizes the reliable gains and the full table appears here.

Where and what we changed. Sec. 4.2.1, the confidence intervals in Table 1, the paired-bootstrap discussion accompanying Table 10, and Table R1 of this letter.

[Manuscript Table 6] 2×2 ambiguity interaction on cross-fitted Intentionomy train predictions. Cells give Δ per-sample F1 against the shared baseline; * marks Holm-corrected $p < 0.05$.

		Low supervisory ($\omega \geq 2/3$)		High supervisory ($\omega = 1/3$)	
		Low sem.	High sem.	Low sem.	High sem.
n		1,271	588	5,099	5,781
Base	F1	72.23	41.56	40.54	22.90
	+SLR-C	-0.43	+4.08*	+0.49	+1.17*
Δ F1	+UTD	-0.05	-0.07	+0.03	+0.84*
	Full FDIL	-0.71	+2.18	+0.15	+1.14*

Comment 3

Moreover, the evaluation is limited to the single benchmark dataset that makes it impossible to assess generalization ability of the approach.

Response: We agree, and the manuscript now carries a cross-dataset study on two label-distribution benchmarks.

Sec. 4.5 evaluates FDIL on Flickr-LDL (11,150 images) and Emotion6 (1,980 images), each on its untouched official test split over five seeds with paired t -tests. These vary a different axis than a further multi-label dataset would. Label distribution learning replaces independent per-class decisions with a point on the probability simplex, so the threshold search disappears and only the two functional roles remain.

Results. On Flickr-LDL, FDIL significantly improves rank correlation and the divisiveness error but significantly degrades Clark distance. On Emotion6

Table R1: Paired t -test of FDIL against each matched-protocol method over five seeds, for all six reported metrics. Δ is the mean per-seed difference in points; CI_{95} is its 95% t -interval; p_H is Holm-corrected across the family of compared methods within each metric. The manuscript summarizes this table in Sec. 4.2.1 and carries the corresponding intervals in Table 1.

Metric	Method	Δ	CI_{95}	t	p	p_H
Macro F1	CLIP ViT-L/14	+4.57	[3.37, 5.76]	10.62	<0.001	0.0014
	UTD only (ours)	+2.00	[0.10, 3.91]	2.92	0.0434	0.0434
	HLEG	+4.52	[3.40, 5.63]	11.24	<0.001	0.0014
	LabCR	+4.03	[2.86, 5.19]	9.60	<0.001	0.0014
	IntCLIP	+6.92	[5.84, 7.99]	17.81	<0.001	<0.001
Micro F1	CLIP ViT-L/14	+4.92	[2.20, 7.64]	5.02	0.0074	0.0148
	UTD only (ours)	+1.88	[0.35, 3.41]	3.42	0.0268	0.0268
	HLEG	+5.05	[3.27, 6.83]	7.89	0.0014	0.0042
	LabCR	+3.12	[2.40, 3.83]	12.13	<0.001	0.0011
	IntCLIP	+5.16	[4.29, 6.03]	16.48	<0.001	<0.001
Samples F1	CLIP ViT-L/14	+4.29	[1.95, 6.64]	5.09	0.0070	0.0141
	UTD only (ours)	+1.60	[-0.26, 3.47]	2.39	0.0755	0.0755
	HLEG	+4.75	[3.18, 6.31]	8.42	0.0011	0.0044
	LabCR	+2.80	[1.54, 4.06]	6.17	0.0035	0.0105
	IntCLIP	+5.30	[4.07, 6.54]	11.91	<0.001	0.0014
Avg. F1	CLIP ViT-L/14	+4.59	[2.88, 6.31]	7.43	0.0018	0.0035
	UTD only (ours)	+1.83	[0.11, 3.55]	2.96	0.0416	0.0416
	HLEG	+4.77	[3.39, 6.16]	9.55	<0.001	0.0020
	LabCR	+3.32	[2.46, 4.17]	10.79	<0.001	0.0017
	IntCLIP	+5.79	[4.86, 6.73]	17.26	<0.001	<0.001
mAP	CLIP ViT-L/14	+3.76	[2.93, 4.59]	12.58	<0.001	<0.001
	UTD only (ours)	+1.64	[0.98, 2.30]	6.94	0.0023	0.0023
	HLEG	+4.11	[3.55, 4.68]	20.22	<0.001	<0.001
	LabCR	+3.41	[2.48, 4.35]	10.16	<0.001	0.0011
	IntCLIP	+7.12	[6.80, 7.44]	61.60	<0.001	<0.001
Hard F1	CLIP ViT-L/14	+7.14	[4.73, 9.56]	8.21	0.0012	0.0036
	UTD only (ours)	+4.15	[0.93, 7.37]	3.58	0.0232	0.0232
	HLEG	+8.47	[6.18, 10.75]	10.28	<0.001	0.0020
	LabCR	+7.67	[4.85, 10.48]	7.56	0.0016	0.0036
	IntCLIP	+10.51	[8.29, 12.73]	13.13	<0.001	0.0010

the only reliable change is Clark distance, the very metric it loses on Flickr-LDL. The margins are small and Emotion6’s 594-image test split has little power at this effect size, so we do not present either dataset as a new state of the art for label distribution learning. The recent LDL-DVS objective [25], run at its default weight on identical features, splits, and seeds, stays essentially identical to the matched baseline under these strong frozen CLIP features, which also limits what that comparison can establish.

What we do take from the study. The division of labour replicates on both datasets. SLR-C moves the pointwise-fit metrics (KLD, cosine, μ) and only those, while gated UTD moves rank correlation and leaves KLD no better. That separation holds where the two modules cannot share a fitted threshold, which is the most plausible way it could otherwise have been an artefact. Table R2 below gives the seed dispersion and per-metric p -values that the manuscript table omits for space.

Where and what we changed. the new Sec. 4.5 with Table 14, the dataset protocols and label-distribution metrics in Sec. 4, and the added transfer limitation in Sec. 5.

Table R2: Seed dispersion and per-metric significance behind manuscript Table 14: official test splits, mean $\pm$ standard deviation over five seeds. Δ and p are for FDIL against the matched baseline (paired t -test over the five paired runs). LDL-DVS [25] is the published divisiveness objective, run on identical features, splits, and seeds. The manuscript table reports the means only, for space.

Metric	Baseline	LDL-DVS	+ SLR-C	+ UTD	FDIL	Δ	p
<i>Flickr-LDL</i>							
Chebyshev $\downarrow$	0.2039 $\pm$ 0.0012	0.2039 $\pm$ 0.0012	0.2031 $\pm$ 0.0006	0.2026 $\pm$ 0.0011	0.2027 $\pm$ 0.0007	-0.0011	0.178
Clark $\downarrow$	2.3033 $\pm$ 0.0044	2.3034 $\pm$ 0.0044	2.3031 $\pm$ 0.0024	2.3123 $\pm$ 0.0025	2.3113 $\pm$ 0.0027	+0.0079	0.008
KLD $\downarrow$	0.3433 $\pm$ 0.0016	0.3433 $\pm$ 0.0016	0.3406 $\pm$ 0.0010	0.3454 $\pm$ 0.0029	0.3433 $\pm$ 0.0016	+0.0000	0.978
Cosine $\uparrow$	0.8966 $\pm$ 0.0007	0.8966 $\pm$ 0.0007	0.8973 $\pm$ 0.0007	0.8961 $\pm$ 0.0013	0.8967 $\pm$ 0.0006	+0.0001	0.874
Spearman $\uparrow$	0.7109 $\pm$ 0.0026	0.7108 $\pm$ 0.0026	0.7129 $\pm$ 0.0010	0.7137 $\pm$ 0.0012	0.7141 $\pm$ 0.0012	+0.0032	0.017
μ (%) $\uparrow$	71.40 $\pm$ 0.14	71.40 $\pm$ 0.14	71.58 $\pm$ 0.09	71.31 $\pm$ 0.18	71.45 $\pm$ 0.11	+0.05	0.542
DVSE $\downarrow$	0.0772 $\pm$ 0.0020	0.0772 $\pm$ 0.0020	0.0764 $\pm$ 0.0014	0.0729 $\pm$ 0.0011	0.0738 $\pm$ 0.0009	-0.0034	0.009
<i>Emotion6</i>							
Chebyshev $\downarrow$	0.1794 $\pm$ 0.0019	0.1794 $\pm$ 0.0019	0.1797 $\pm$ 0.0016	0.1804 $\pm$ 0.0018	0.1802 $\pm$ 0.0011	+0.0008	0.438
Clark $\downarrow$	1.5979 $\pm$ 0.0025	1.5981 $\pm$ 0.0026	1.5976 $\pm$ 0.0053	1.5949 $\pm$ 0.0047	1.5910 $\pm$ 0.0023	-0.0069	0.013
KLD $\downarrow$	0.2532 $\pm$ 0.0026	0.2532 $\pm$ 0.0026	0.2515 $\pm$ 0.0019	0.2537 $\pm$ 0.0033	0.2513 $\pm$ 0.0006	-0.0019	0.190
Cosine $\uparrow$	0.8947 $\pm$ 0.0014	0.8946 $\pm$ 0.0014	0.8955 $\pm$ 0.0013	0.8941 $\pm$ 0.0018	0.8953 $\pm$ 0.0008	+0.0006	0.439
Spearman $\uparrow$	0.7223 $\pm$ 0.0025	0.7222 $\pm$ 0.0024	0.7231 $\pm$ 0.0026	0.7256 $\pm$ 0.0026	0.7219 $\pm$ 0.0026	-0.0003	0.740
μ (%) $\uparrow$	64.20 $\pm$ 0.37	64.20 $\pm$ 0.37	64.52 $\pm$ 0.28	64.12 $\pm$ 0.43	64.41 $\pm$ 0.12	+0.21	0.304
DVSE $\downarrow$	0.0784 $\pm$ 0.0009	0.0783 $\pm$ 0.0009	0.0781 $\pm$ 0.0006	0.0791 $\pm$ 0.0014	0.0781 $\pm$ 0.0009	-0.0003	0.683

Comment 4

The analysis of computational efficiency, visualization of failures, and discussion of deployment are needed to make the contribution stronger.

Response: All three are in the manuscript, and we reproduce the efficiency table and the failure case below.

Efficiency. Table 15 reports inference-active parameters, per-image FLOPs, and batch-1 latency under one shared protocol, at 2.41M parameters and 0.47 ms per image for full FDIL, with UTD-only at exactly baseline cost.

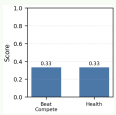
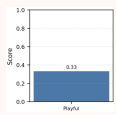
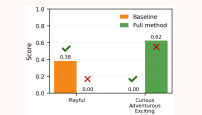
Failure visualization. Fig. 5, reproduced below, shows both a success and a genuine failure case, and the interpretability analysis reports that only two of the 28 classes regress. The per-class attribution behind that sentence was condensed out of the manuscript for space, so we reproduce it here (Table R3). The sole non-trivial regression is *CreativeUnique* (-5.3), a broad category overlapping several neighbouring design- and art-related intents that the residual student occasionally resolves toward a competing neighbour.

Deployment. The efficiency discussion notes that the UTD branch can be deployed alone at zero additional inference cost, since the text teacher is discarded after training, and the Limitations paragraph records the offline text-generation cost any deployment must budget for.

Where and what we changed. Table 15 and its surrounding discussion, the Interpretability Analysis subsection and Fig. 5, and the deployment and preparation-cost notes in Sec. 5.

[Manuscript Table 15] Feature-level inference cost on frozen CLIP ViT-L/14 features: params (inference-active heads), per-image FLOPs, and batch-1 latency on an RTX 4090. HVU-CLIP releases no code and cannot be measured; IntentMLM and CoT4Intent run MLLM inference pipelines whose cost is orders of magnitude higher.

Method	Params (M)	FLOPs (M)	Latency (ms)
Baseline	1.20	2.40	0.109
FDIL (UTD only)	1.20	2.40	0.109
FDIL	2.41	4.93	0.466
HLEG	203.70	12 075.9	2.725
IntCLIP	0.02	22.02	0.111

Input	Soft Probability	Predictions	Rationale
			The image is static and focused on the basketball hoop, which does not convey a sense of adventure or excitement. The lack of movement or other elements that might suggest exploration or discovery further disambiguates this intent. The image does not show any elements that suggest curiosity, adventure, or excitement. There are no signs of exploration, discovery, or dynamic action that would typically be associated with these intents.
			The image shows a card with the text "STAY CURIOUS THINK GOOD" prominently displayed. The card is placed on a plain, light-colored surface, likely a bed or a table. The text on the card is motivational and encourages positive thinking, which aligns more with a reflective or aspirational intent.

[Manuscript Fig. 5] Qualitative case study on representative Intentionomy samples. Top: the baseline predicts an incorrect alternative and FDIL rejects that distractor using counterfactual rationale evidence. Bottom: a genuine failure of FDIL, on a label with only 0.33 annotator agreement, where its alternative remains semantically plausible.

Table R3: Per-class F1 attribution of FDIL versus the CLIP baseline (matched class-wise protocol, representative seed). Top: five largest improvements. Bottom: the only two regressions among all 28 classes. This table was condensed out of the manuscript for the page limit; the manuscript retains its one-sentence summary in the Interpretability Analysis.

Intent class	Subset	Baseline	FDIL	Δ
WorkILike	Hard	14.6	43.0	+28.5
PassionAboutSomething	Hard	11.3	28.6	+17.3
SocialLifeFriendship	Hard	33.9	51.2	+17.2
ManageableMakePlan	Medium	17.5	32.0	+14.5
BeatCompete	Medium	54.6	68.1	+13.5
FineDesignLearnArt-Arch	Easy	84.0	83.1	-1.0
CreativeUnique	Medium	41.9	36.6	-5.3

3. Response to Reviewer 3 (Associate Editor)

We thank the Associate Editor for clarifying the intent of the previous comment. The Abstract is now built around comparative statements, as requested.

Comment 1

Although the manuscript has been revised in response to the previous reviewer comments, the Abstract still does not fully address the main concern. My previous comment was not intended to encourage the authors to simply present more numerical values. Rather, I expected a quantitative comparison with existing methods, such as clearly stating the percentage improvement over the baseline. Presenting objective comparative metrics is essential for highlighting the significance of the proposed method and attracting readers' interest. Therefore, I believe this part still requires further revision.

Response: *The Abstract now states relative gains over the strongest prior result rather than absolute scores.*

We agree that a relative comparison conveys the size of the advance in a way absolute scores do not. The Abstract now states an explicit relative gain over the strongest existing result in Table 1, and reads "On Intentionomy, FDIL leads the state-of-the-art by 6.1% relative average F1 and 28.0% on the hard subset."

Both percentages are computed directly from the main comparison table, and together they identify the reference class (the prior state of the art) and the two aspects of performance improved (overall accuracy and difficult examples).

The surrounding text then summarizes the cross-dataset evidence on Flickr-LDL and Emotion6 without adding a further list of isolated metric values. This keeps the Abstract focused on objective comparative gains, as requested, and leaves the complete per-method results and five-seed confidence intervals to Table 1.

Where and what we changed. the rewritten quantitative-comparison sentences in the Abstract.

4. Response to Reviewer 4

We thank the reviewer for the continued scrutiny. The dataset-scale and recent-baseline requests are answered with new experiments, the theoretical request with a rebuilt analysis in the information-theoretic and feature-selection terms the reviewer names, and the proofreading item with a bibliography-wide pass.

Comment 1

There must be larger-scale datasets to better evaluate the proposed methods.

Response: *We agree, and the cross-dataset evidence has moved from the letter into the manuscript as a full subsection.*

Sec. 4.5 evaluates FDIL on Flickr-LDL (11,150 images, per-image annotator vote distributions over eight emotions) and Emotion6 (1,980 images, seven categories including neutral). Each uses its untouched official test split, five seeds with paired t -tests, and out-of-fold teacher predictions so no training target derives from an evaluated sample. We chose these to vary a different axis rather than simply to add images, because label distribution learning removes the class-wise threshold search entirely, leaving only the two functional roles to carry the transfer.

Results. On Flickr-LDL, FDIL significantly improves rank correlation and the divisiveness error but significantly degrades Clark distance. On Emotion6 the only reliable change is Clark distance, the metric it loses on Flickr-LDL. We do not aggregate the two into a single favourable summary. What replicates on both is the division of labour the decoupling predicts, with SLR-C moving the pointwise-fit metrics and gated UTD moving rank correlation, in a setting where the two cannot share a fitted threshold.

For completeness, Table R2 of this letter gives the seed dispersion and per-metric p -values omitted from the manuscript table for space, including the matched LDL-DVS [25] comparison, which stays essentially identical to the baseline under these strong frozen CLIP features.

Where and what we changed. the new Sec. 4.5 with Table 14, the dataset protocols and label-distribution metrics in Sec. 4, and the added transfer limitation in Sec. 5.

Comment 2

More comparisons with state-of-the-art Large Language Model methods (such as 2026) are needed.

Response: *We agree, and added two 2026 intent-recognition methods plus four recent MLLMs, and both tables are reproduced below.*

CoT4Intent (2026) probes hallucination when MLLMs are prompted with chain-of-thought reasoning for visual intention, and is now a row in Table 1 (49.54 Avg. F1). It bears directly on our design rationale, because it finds that MLLM-generated intent reasoning contains evidence that does not ground the predicted intent, which is why UTD consumes rationales through an agreement gate rather than trusting them uniformly. We make this connection explicit in Related Work.

HVU-CLIP (TPAMI 2026) is the journal extension of IntCLIP by the same authors. Its published numbers (50.22 Avg. F1) enter Table 1 under its own CLIP ViT-B/32 backbone and are marked accordingly, since no code has been released. Because that backbone differs from ours, Table 13 adds a same-backbone comparison under ResNet101, where FDIL is best on four of the five reported columns, exceeding HVU-CLIP on Macro F1 (43.20 vs. 41.24), Samples F1 (52.02 vs. 50.41), and Avg. F1 (48.62 vs. 47.96), while Micro F1 is lower (50.63

vs. 52.22).

Zero-shot MLLMs. The zero-shot comparison now has its own table (Table 2), adding three 2026 open-weight models, namely Qwen3.5-9B, Step3-VL-10B, and Gemma-4-E4B-IT, plus GPT-5.6 Sol. The frontier model is genuinely strong, since its 57.50 mAP and 36.19 hard-subset F1 exceed both task-trained references, while its 49.04 Avg. F1 stays 8.24 points below FDIL at billion-parameter inference cost. We read this as stronger pretrained priors helping ranking and ambiguity-prone classes, with task-specific learning retaining a clear advantage on aggregate fixed-threshold F1.

Where and what we changed. the two new works discussed individually in Sec. 2, the CoT4Intent and HVU-CLIP rows in Table 1, the same-ResNet101 comparison in Table 13, and the dedicated zero-shot comparison in Table 2.

[Manuscript Table 1] Main comparison on Intentionomy test split (excerpt: recent methods and ours). Avg. is the mean of Macro/Micro/Samples F1. [†]quoted under its own CLIP ViT-B/32 backbone because no code has been released; a backbone-matched comparison under ResNet101 is given in manuscript Table 13.

Method	Publication	Macro F1	Micro F1	Samples F1	Avg.
PIP-Net	TMM 2023	32.57	45.94	47.05	41.85
CoT4Intent	-	41.70	53.10	53.82	49.54
HVU-CLIP (ViT-B/32) [†]	TPAMI 2026	42.91	55.22	52.54	50.22
LabCR (CLIP ViT-L/14)	TIP 2024	47.87 $\pm$ 0.38	56.78 $\pm$ 0.97	57.23 $\pm$ 1.25	53.96 $\pm$ 0.71
HLEG (CLIP ViT-L/14)	TAC 2023	47.38 $\pm$ 0.75	54.85 $\pm$ 1.70	55.29 $\pm$ 1.06	52.50 $\pm$ 0.98
IntCLIP (CLIP ViT-L/14)	ECCV 2024	44.98 $\pm$ 1.12	54.73 $\pm$ 0.49	54.73 $\pm$ 1.18	51.48 $\pm$ 0.82
IntentMLM	PR 2026	43.05	54.77	56.75	51.52
CLIP ViT-L/14	ICML 2021	47.33 $\pm$ 1.15	54.98 $\pm$ 2.90	55.74 $\pm$ 2.64	52.68 $\pm$ 1.96
FDIL (UTD only)	Ours	49.90 $\pm$ 1.18	58.01 $\pm$ 1.03	58.43 $\pm$ 1.13	55.45 $\pm$ 1.02
FDIL	Ours	51.90 $\pm$ 0.95	59.90 $\pm$ 0.62	60.03 $\pm$ 0.94	57.28 $\pm$ 0.75

[Manuscript Table 2] Zero-shot MLLM comparison on the Intentionomy test split, same 28-label set and single shared prompt. *Hard* is F1 on the hard difficulty subset. Zero-shot rows are single deterministic passes with a fixed 0.5 threshold; GPT-5.6 Sol is evaluated with reasoning disabled. [‡]published result quoted from IntentMLM; per-class scores unavailable.

Model	Macro	Micro	Samples	Avg.	mAP	Hard
Zero-shot prompting						
Qwen3.5-9B	37.11	36.50	36.40	36.67	42.99	22.78
Step3-VL-10B	38.21	40.21	36.10	38.17	42.58	26.64
Qwen3-VL-8B-Instruct	38.51	40.75	39.63	39.63	38.84	28.37
InternVL3-8B	37.54	41.85	40.50	39.97	37.67	25.51
Gemma-4-E4B-IT	38.76	42.44	39.25	40.15	40.42	27.08
GPT-4o [‡]	39.48	40.57	41.45	40.50	-	-
GPT-5.6 Sol	47.58	49.64	49.91	49.04	57.50	36.19
Task-trained reference						
CLIP ViT-L/14	47.33	54.98	55.74	52.68	50.67	28.45
FDIL (ours)	51.90	59.90	60.03	57.28	55.10	35.02

Comment 3

The paper lacks theoretical analysis or deeper learning insights. It must add more value to the paper if more theoretical analysis were conducted, e.g., from information-theoretical or feature selection perspectives.

Response: *We agree, and rebuilt Sec. 3.5 around exactly those two perspectives, with Sec. 4.3.2 testing what they predict.*

Information-theoretic side. Eq. (15) treats the intent label as a random human judgement rather than a deterministic ground truth, giving the exact identity $H(Y_c | Z) = H(Y_c | X) + I(Y_c; X | Z)$, which adds irreducible subjective disagreement to the intent-relevant information the representation discards, the latter unrecoverable downstream by the data-processing inequality. SLR-C’s contribution is then stated as a conditional-information claim, $H(Y_c | B, S) = H(Y_c | B) - I(Y_c; S | B)$ (Eq. (16)), where the gain is exactly the label-relevant information in the prior that the baseline prediction does not already carry, and vanishes when the prior is redundant. That is why SLR-C supplements under-used semantic information rather than offering a second opinion on the same evidence, and the required non-redundancy is visible in the prior ablation (Table 7), where no single source dominates.

Feature-selection side. Eq. (17) formalizes the Top- K stage as local hypothesis screening followed by margin correction. Inside the screened set, reranking shifts a confusable pair’s margin by exactly $\alpha(\tilde{s}_{c+}^{\text{prior}} - \tilde{s}_{c-}^{\text{prior}})$, and nothing outside it changes. In screening terms K trades coverage against distractor noise, and both ends are measured. The baseline Top-10 set covers 83.9% of positive labels, and the sensitivity study (Table 8) shows an interior optimum at $K = 10$.

UTD as reliability-weighted privileged information. The rationale teacher is now stated for what it is, a channel available in training and discarded at test time [27, 28]. That turns the gate into an estimation question with an answer, because the mean-squared-error-optimal weight $g^* = \sigma_v^2(\omega) / (\sigma_v^2(\omega) + \sigma_t^2 + b_t^2)$ of Eq. (18) is monotonically decreasing in agreement, so our $g = 1 - \omega$ is a parameter-free monotone approximation of it rather than an unstructured heuristic.

The experiment these derivations predict. Sec. 4.3.2 stratifies samples by the two ambiguities and measures each module per cell, cross-fitted over the train split because it is the only split carrying per-sample agreement. The semantic axis confirms the prediction cleanly. The supervisory axis is directionally consistent but weaker, and its low-ambiguity stratum holds only 1,859 images, so we claim no monotone trend. The full model also does not exceed its stronger component in the doubly-ambiguous cell. We therefore use the experiment to support the predicted localization of the two effects, not to claim a gain from composing them.

Where and what we changed. the derivations (15)–(19) and the module mapping in Sec. 3.5, the interaction experiment in Sec. 4.3.2 and Table 6, and the cross-dataset test in Sec. 4.5.

Comment 4

The overall framework is over-complicated and lacks clarity in design justification. Multiple modules are introduced, but their individual contributions and interactions are not clearly disentangled. The ablation studies, while present, are not sufficient to demonstrate that each component is necessary rather than redundant, making the method appear over-engineered.

Response: *FDIL has two functional modules, only one of which exists at inference, and four independent tests separate necessity from redundancy, two of them reproduced below.*

On complexity. There are two modules, not many, namely a fixed semantic prior with a residual head (SLR-C) and a training-only text teacher (UTD). At inference UTD contributes nothing, because the teacher is discarded, so the deployed model is a lightweight residual MLP over frozen features at 2.41M parameters and 0.47 ms per image (Table 15).

On necessity rather than redundancy. The manuscript offers four independent tests. The unified control shows that merging both signals into one distillation target performs no better than either alone (Table 5). Shuffling rationales across images is worse than removing UTD entirely (-10.40 against -4.47 Avg. F1), so the gain depends on image-specific content rather than on adding external text (Table 12). Removing or flattening the agreement gate degrades results consistently (-1.97 and -1.12 Avg. F1). And the two label-distribution transfers separate the modules by which metric each one moves, on both datasets, where they cannot share a fitted threshold. Redundant components do not act on different quantities.

Where a component is not clearly necessary. FDIL’s advantage over its own UTD-only ablation does not reach significance on Samples F1 at $n = 5$ ($p_H = 0.0755$ in Table R1), although it does on Macro, Micro, Avg., mAP, and hard-subset F1. The case for the decoupled design therefore rests on the overall evidence together with modularity, since each function stays separately measurable and UTD can be deployed alone at zero additional inference cost.

Presentation. The unified-control and interaction experiments are now one subsection arguing effectiveness and necessity together, and every results subsection has been reduced to its table or figure, a short analysis, and a conclusion.

One note on notation. The notation table added in the previous round has been withdrawn, purely because the page limit under double spacing left no room once the new theoretical section, the interaction experiment, and the cross-dataset study entered the manuscript. No symbol became less explicit, because each is introduced in words at its first occurrence, including those the table used to carry, and we would gladly reinstate it if the editor can grant the additional page.

Where and what we changed. Tables 5, 6, 12, and 4, the dissociation in Sec. 4.5, the significance discussion in Sec. 4.2.1, and the modularity argument

in Sec. 5.

[Manuscript Table 5] Unified versus functionally decoupled ambiguity handling. Results are means over three seeds.

Method	Macro	Micro	Samples	Avg.	Hard
Unified UTD only	50.20	58.03	58.79	55.68	30.27
Unified SLR-C only	51.29	60.10	59.22	56.87	34.01
Unified joint-target KD	50.69	58.35	57.96	55.66	32.41
FDIL	52.13	60.06	60.28	57.49	35.02

[Manuscript Table 12] Negative controls on the UTD text teacher (Δ in points; negative denotes degradation).

Condition	Macro	Micro	Samples	Avg.	mAP	Hard
FDIL (full)	52.13	60.06	60.28	57.49	55.10	35.02
w/o UTD	-4.23	-4.08	-5.10	-4.47	-4.93	-5.31
Shuffled rationale	-7.75	-12.39	-11.01	-10.40	-9.04	-7.58
Ungated teacher	-2.44	-2.16	-1.30	-1.97	-1.49	-3.33
Uniform agreement gate	-1.16	-0.93	-1.21	-1.12	-1.33	+1.13

Comment 5

The manuscript must require careful proofreading, such as [7] The manuscript must require careful proofreading and [9] Computer Vision - ECCV 2024.

Response: *Both entries are corrected, followed by a bibliography-wide pass.*

Entry [9] now reads *Computer Vision – ECCV 2024* with a proper en dash instead of the stray hyphen, and we re-checked entry [7] together with all neighbouring entries. We then standardized venue names and abbreviations across the bibliography and verified that every entry carries a complete venue field, applying the same convention to all newly added references. We also re-read the manuscript for grammar, acronym consistency, and cross-reference correctness.

Where and what we changed. the corrected venue fields in the Bibliography including entry [9], the standardized venue names and abbreviations throughout, and the terminology and cross-reference corrections in the manuscript body.

5. Response to Reviewer 6

We thank the reviewer for the positive assessment of the writing and structure, and for the precise methodological requirement, which we have implemented in full.

Comment 1

The paper presents a functionally decoupled framework for visual-intent recognition that addresses supervisory and semantic ambiguity using structured semantic priors and uncertainty-aware distillation. The manuscript is well written, clearly structured, and the proposed contribution is reasonably good and relevant to the field. However, the performance improvement over comparable state-of-the-art methods is relatively small, which limits the strength of the empirical contribution. Although the authors report results using mean $\pm$ standard deviation over three runs, this alone is not sufficiently convincing for establishing statistical superiority. An explicit paired t-test, applied consistently against the strongest state-of-the-art methods, should therefore be provided to demonstrate that the observed improvements are statistically significant rather than caused by random variation.

Response: *We implemented exactly the requested test, namely five seeds with paired two-sided t-tests against every compared method, Holm-corrected. Table R1 of this letter gives the full statistics.*

We agree that mean $\pm$ std over three runs does not establish statistical superiority.

Protocol. We extend from three to **five** independent seeds. For each seed we retrain, on that same seed, the matched-protocol reproductions of the three strongest comparable methods, namely HLEG, LabCR, and IntCLIP, over identical frozen CLIP ViT-L/14 features, the identical split, and identical validation-only class-wise thresholding. This gives five *paired* observations per method, with the training run as the pairing unit, so a paired two-sided *t*-test on the per-seed differences is well defined. Since FDIL is tested against five methods per metric, we also report Holm-corrected *p*-values.

Outcome. Avg. F1, mAP, and hard-subset F1 are significant against every compared method after correction. Against the strongest compared method, LabCR, the gains are +3.32 Avg. F1 and +7.67 hard-subset F1. The class-wise-thresholded F1 metrics carry more seed-to-seed variance because thresholds are fitted on a 498-image validation split, and the sole non-significant contrast is Samples F1 against our own UTD-only ablation ($p_H = 0.0755$), which we report alongside the significant ones. This establishes statistical superiority over the matched published methods while remaining consistent with the modest effect size the reviewer correctly identified. Sec. 4.2.1 states the outcome and protocol, and the per-method table is in this letter because the manuscript is at the 35-page limit.

One note on the table. The matched-protocol rows of Table 1 were regenerated together with the significance test, so that the reported values and the test derive from a single reproducible run. They differ from the previous revision by a few tenths of a point.

Where and what we changed. Sec. 4.2.1, the confidence intervals and regenerated matched-protocol rows in Table 1 whose caption now states five seeds,

and Table R1 of this letter.

6. Response to Editor-in-Chief Checklist

We checked the revision against each item and report the outcome below.

Comment 1

Take a careful look at your bibliography and they cover the state of the art. Missing references from last and current year most probably would mean you are missing the state of the art and the revision process can be delayed being asked to update it. Please do not make excessive citation to arXiv papers, but substitute them with their peer-reviewed versions, or papers from a single conference series. Do not cite large groups of papers without individually commenting on them. So we discourage “In prior work [1,2,3,4,5,6] ...”. Your bibliography in the final version after the revision still should be between 35-55 items.

Response: *Recent coverage extended, all new works discussed individually, 51 cited items.*

Recent coverage. We add two 2026 works, CoT4Intent and HVU-CLIP (TPAMI 2026), alongside the IntCLIP (ECCV 2024) and IntentMLM (Pattern Recognition 2026) references already present, plus the recent label-distribution-learning line (LDL-DVS, ICML 2026). The bibliography now covers the current state of the art on this benchmark.

Preprint citations. We checked every entry for a peer-reviewed version and cite it wherever one exists. The few remaining preprint citations are model technical reports and system cards for the MLLMs we evaluate zero-shot, which have no peer-reviewed version to substitute.

No unsupported citation groups. We reviewed the citation sites in the revised material. The newly added intent-recognition and label-distribution methods are each discussed by name and contribution in Sec. 2, rather than appended to an unexplained bracket list.

Bibliography size. The compiled revised manuscript contains 51 cited items, within the required 35–55 range.

Where and what we changed. the new entries and corrected venue fields in the Bibliography, and the individually discussed additions in Sec. 2.

Comment 2

Please make sure the revised version is relevant to the readership of the Pattern Recognition field. To this end, please make sure you cite RECENT work from the field of pattern recognition not only the Pattern Recognition journal.

Response: *The bibliography spans the recent pattern-recognition literature rather than a single venue.*

Cited venues include IEEE TPAMI (HVU-CLIP, 2026), Pattern Recognition (IntentMLM, 2026), ICML 2026 (LDL-DVS), ICML 2025 (δ -LDL), IEEE TKDE (LDL-LRR, 2023), IEEE TNNLS (LDL-DPA, 2024), IEEE TIP (LabCR 2024, CPAD 2023), IEEE TAC (HLEG 2023), IEEE TMM (PIP-Net 2023), and ECCV 2024 (IntCLIP), together with the multi-label classification and noisy-label lines from CVPR, NeurIPS, ICML, ICLR, and AAAI.

The problem itself, multi-label recognition under ambiguous and subjective supervision with calibration and threshold selection treated explicitly, sits squarely within the journal’s scope.

Where and what we changed. the venue distribution of the Bibliography, and the multi-label and label-noise method families compared in Table 1.

Comment 3

Although the revision could lead to extending your article, it still can not exceed the page limits or violate the format, i.e. double spaced SINGLE column with a maximum of 35 pages for a regular paper and 40 pages for a review.

Response: *The compiled marked-up manuscript is **35 pages**, double-spaced and single-column, exactly at the limit.*

The revised manuscript is double-spaced and single-column throughout, in the `elsarticle review` format, with no reduction of line spacing anywhere, neither in the body nor the tables nor the bibliography.

To fit the reviewer-requested statistical validation, theoretical analysis, zero-shot MLLM comparison, interaction experiment, and cross-dataset generalization study within that limit, we condensed repeated protocol text and kept the two label-distribution benchmarks in one combined table. An earlier exploratory EMOTIC transfer study and several supplementary diagnostic tables are not included in the submitted manuscript. The per-class attribution table among them is reproduced in this letter (Table R3). These choices preserve the requested evidence while keeping the article within both the page and formatting requirements.

Where and what we changed. the 35-page compiled review manuscript and its unchanged double-spaced, single-column format.

7. Closing Statement

We thank the Associate Editor and all reviewers for a demanding and constructive second round. The two concerns the Associate Editor emphasized, statistical validation and evaluation breadth, drove substantive new work rather than rewriting. We now report five-seed training with confidence intervals and paired t -tests against every compared method, together with a cross-dataset study on the Flickr-LDL and Emotion6 label-distribution benchmarks that is placed in the manuscript itself.

Three results are worth stating plainly.

- The cross-dataset study reports a significant *degradation* in Clark distance on Flickr-LDL, and on Emotion6 only the Clark-distance improvement is reliable, the very metric FDIL loses on Flickr-LDL. The manuscript states that the two datasets disagree on which metric improves, and confines the claim there to the division of labour between the modules.
- The matched LDL-DVS objective stays essentially identical to the baseline under our frozen features, which limits what that comparison can establish.
- The significance analysis reports the one contrast that does not reach significance alongside those that do, namely Samples F1 against our own UTD-only ablation.

The submitted marked-up manuscript remains double-spaced and single-column and compiles to exactly **35 pages**, achieved by condensing repeated protocol text rather than by reducing line spacing. The three supplementary tables that did not fit are reproduced in this letter, alongside the manuscript tables and the case-study figure the reviewers asked about.

We believe the revision now meets the standards the reviewers set, and we are grateful for the care that produced these improvements.